

## 3.3 Kinematics in 2 Dimensions

Mathematics B (MEI) — OCR B — A-Level (H640) — Mechanics — spec refs k9–k11

**What this topic covers.** Extending kinematics to two dimensions using vectors — position, velocity and acceleration as vector functions of time, the vector forms of the constant acceleration formulae, relative position and velocity, and finding the Cartesian equation of a path.

### 1. Vector kinematics

In two dimensions each quantity becomes a vector with **i** and **j** components (spec ref k9).

| Quantity     | Symbol   | Relationship                          |
|--------------|----------|---------------------------------------|
| Position     | <b>r</b> | Measured from a fixed origin          |
| Velocity     | <b>v</b> | $\mathbf{v} = \frac{d\mathbf{r}}{dt}$ |
| Acceleration | <b>a</b> | $\mathbf{a} = \frac{d\mathbf{v}}{dt}$ |

Going the other way,  $\mathbf{v} = \int \mathbf{a} dt$  and  $\mathbf{r} = \int \mathbf{v} dt$ , each with a vector constant of integration fixed by the initial conditions.

Differentiation and integration act **component by component**, which is what makes two-dimensional motion tractable: the **i** and **j** directions behave independently.

**Worked example.** A particle has  $\mathbf{r} = 3t\mathbf{i} + t^2\mathbf{j}$ . Find its velocity, speed at  $t = 2$ , and acceleration.

$$\mathbf{v} = 3\mathbf{i} + 2t\mathbf{j}$$

At  $t = 2$ ,  $\mathbf{v} = 3\mathbf{i} + 4\mathbf{j}$ , so the speed is  $|\mathbf{v}| = \sqrt{9 + 16} = 5$ .

$$\mathbf{a} = 2\mathbf{j}$$

**Worked example.** A particle has  $\mathbf{a} = 2\mathbf{i} + 6t\mathbf{j}$  and  $\mathbf{v} = \mathbf{i}$  at  $t = 0$ . Find  $\mathbf{v}$  at time  $t$ .

$$\mathbf{v} = \int (2\mathbf{i} + 6t\mathbf{j}) dt = 2t\mathbf{i} + 3t^2\mathbf{j} + \mathbf{c}$$

At  $t = 0$ ,  $\mathbf{v} = \mathbf{i}$ , so  $\mathbf{c} = \mathbf{i}$  and

$$\mathbf{v} = (1 + 2t)\mathbf{i} + 3t^2\mathbf{j}$$

**Speed** is the magnitude of the velocity vector, so it is a scalar:  $|\mathbf{v}| = \sqrt{v_1^2 + v_2^2}$ . Quoting a "speed" as a vector loses the mark.

## 2. Distance and displacement in two dimensions

- **Distance from** the origin at time  $t$  is  $|\mathbf{r}|$ .
- **Displacement** between two times is the difference of the position vectors.
- **Distance travelled** is the length of the path, generally greater than the magnitude of the displacement.

**Worked example.** For  $\mathbf{r} = 2t\mathbf{i} + t^2\mathbf{j}$ , find the average velocity between  $t = 0$  and  $t = 2$ .

Displacement  $= (4\mathbf{i} + 4\mathbf{j}) - \mathbf{0} = 4\mathbf{i} + 4\mathbf{j}$ , over 2 seconds:

$$\text{average velocity} = 2\mathbf{i} + 2\mathbf{j}$$

## 3. Constant acceleration in vector form

The suvat formulae carry over directly, with vectors in place of scalars (k10):

$$\mathbf{v} = \mathbf{u} + \mathbf{a}t$$

$$\mathbf{s} = \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$$

$$\mathbf{s} = \mathbf{v}t - \frac{1}{2}\mathbf{a}t^2$$

$$\mathbf{s} = \frac{1}{2}(\mathbf{u} + \mathbf{v})t$$

**Worked example.** A particle has  $\mathbf{u} = 2\mathbf{i}$  and  $\mathbf{a} = \mathbf{j}$ . Find its displacement after 4 s.

$$\mathbf{s} = 2\mathbf{i}(4) + \frac{1}{2}\mathbf{j}(16) = 8\mathbf{i} + 8\mathbf{j}$$

**Worked example.** A particle starts at  $2\mathbf{i} + \mathbf{j}$  with constant velocity  $3\mathbf{i} - \mathbf{j}$ . Find its position at  $t = 4$ .

$$\mathbf{r} = (2\mathbf{i} + \mathbf{j}) + 4(3\mathbf{i} - \mathbf{j}) = 14\mathbf{i} - 3\mathbf{j}$$

Because  $\mathbf{v} = \mathbf{u} + \mathbf{a}t$ , the **change** in velocity is always parallel to the acceleration. With zero acceleration the velocity is constant and the path is a straight line.

## 4. Relative position and relative velocity

position of  $B$  relative to  $A = \mathbf{r}_B - \mathbf{r}_A$

velocity of  $A$  relative to  $B = \mathbf{v}_A - \mathbf{v}_B$

**Worked example.**  $A$  is at  $3\mathbf{i} + 4\mathbf{j}$  and  $B$  at  $7\mathbf{i} + \mathbf{j}$ .

Position of  $B$  relative to  $A$  is  $4\mathbf{i} - 3\mathbf{j}$ , and the distance between them is  $\sqrt{16 + 9} = 5$ .

**Worked example.**  $A$  has velocity  $5\mathbf{i} + 2\mathbf{j}$  and  $B$  has  $3\mathbf{i} - \mathbf{j}$ . The velocity of  $A$  relative to  $B$  is

$$(5 - 3)\mathbf{i} + (2 - (-1))\mathbf{j} = 2\mathbf{i} + 3\mathbf{j}$$

## Collisions

Two particles **collide** when their position vectors are equal *at the same instant* — that is, both components agree for a single common value of  $t$ .

**Worked example.**  $A$  has  $\mathbf{r}_A = (t + 1)\mathbf{i} + 2t\mathbf{j}$  and  $B$  has  $\mathbf{r}_B = (3t - 1)\mathbf{i} + (t + 1)\mathbf{j}$ . Do they collide?

Equating  $\mathbf{i}$  components:  $t + 1 = 3t - 1$ , so  $t = 1$ .

Equating  $\mathbf{j}$  components:  $2t = t + 1$ , so  $t = 1$ .

Both give the same time, so they collide at  $t = 1$ , at the point  $2\mathbf{i} + 2\mathbf{j}$ .

You must check **both** components and confirm they give the *same*  $t$ . If the two components give different times, the paths cross but the particles are never there together — no collision.

The distance between two particles is least when the magnitude of the relative position vector is minimised, which is found by writing that magnitude as a function of  $t$  and minimising it.

## 5. The Cartesian equation of the path

Eliminate  $t$  between the component equations (k11).

**Worked example.**  $\mathbf{r} = 3t\mathbf{i} + t^2\mathbf{j}$ .

Here  $x = 3t$  and  $y = t^2$ , so  $t = \frac{x}{3}$  and

$$y = \frac{x^2}{9}$$

**Worked example.**  $\mathbf{r} = 4t\mathbf{i} + (3t - 5t^2)\mathbf{j}$ .

$x = 4t$  gives  $t = \frac{x}{4}$ , so

$$y = \frac{3x}{4} - \frac{5x^2}{16}$$

a parabola — the signature of motion under constant acceleration not parallel to the initial velocity.

**Worked example.** For  $\mathbf{r} = (t - 2)\mathbf{i} + (t^2 - 4)\mathbf{j}$ , when does the particle pass through the origin? The  $\mathbf{i}$  component vanishes at  $t = 2$ ; the  $\mathbf{j}$  component vanishes at  $t = \pm 2$ . Both are zero at  $t = 2$ , so it passes through the origin then.

## 6. Common pitfalls

- Reporting a speed as a vector rather than its magnitude.
- Omitting the vector constant of integration.
- Subtracting relative position or velocity the wrong way round.
- Concluding a collision from the  $\mathbf{i}$  components alone.
- Confusing distance from the origin with distance travelled.
- Applying constant-acceleration formulae when the acceleration is a function of  $t$ .
- Eliminating  $t$  from the harder component and introducing a spurious square root.